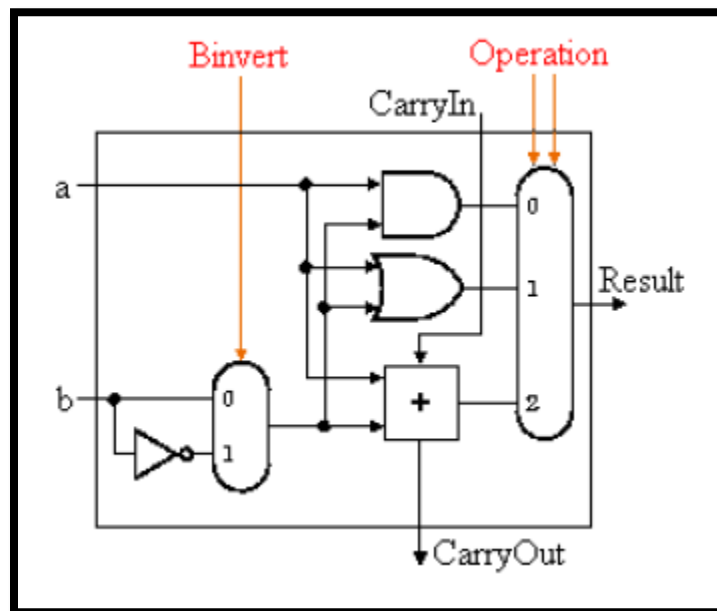


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Q1) Complete the Following Table with Control Lines to Achieve this 4 Operation:



Control Lines	Operation
110	SUB
010	ADD
000	AND
001	OR

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Q2) Express the Following HexaDecimal to Decimal Using IEEE 754 Floating Point Representation:

0x40F00000	0xE4F01100
0100 0000 1111 0000 0000 0000 0000 0000 S E Significant 0 10000001 11100000000000000000000000000000 $=(-1)^S * (1 + \text{Significant}) * 2^{E-127}$ = 7.5	0100 0000 1111 0000 0000 0000 0000 0000 S E Significant 0 10000001 11100000000000000000000000000000 $=(-1)^S * (1 + \text{Significant}) * 2^{E-127}$ = -3.54 * 10 ²²

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Q3) Based on The Final Division Algorithm, Complete 0111(two)/0011(two) by Filling in The Following Table:

Iteration	Divisor (D)	Reminder (R)	Remark
0	0011	0000 0111 0000 1110	Initial state $R = R \ll 1, R_0=0$
1		1101 1110 0000 1110 0001 1100	$R(\text{left}) = R(\text{left}) - D$ Undo $R = R \ll 1, R_0=0$
2		1110 1100 0001 1100 0011 1000	$R(\text{left}) = R(\text{left}) - D$ Undo $R = R \ll 1, R_0=0$
3		0000 1000 0001 0001	$R(\text{left}) = R(\text{left}) - D$ $R = R \ll 1, R_0=1$
4		1110 0001 0001 0001 0010 0010	$R(\text{left}) = R(\text{left}) - D$ Undo $R = R \ll 1$
extra		0001 0010	$R(\text{left}) = R(\text{left}) \gg 1$

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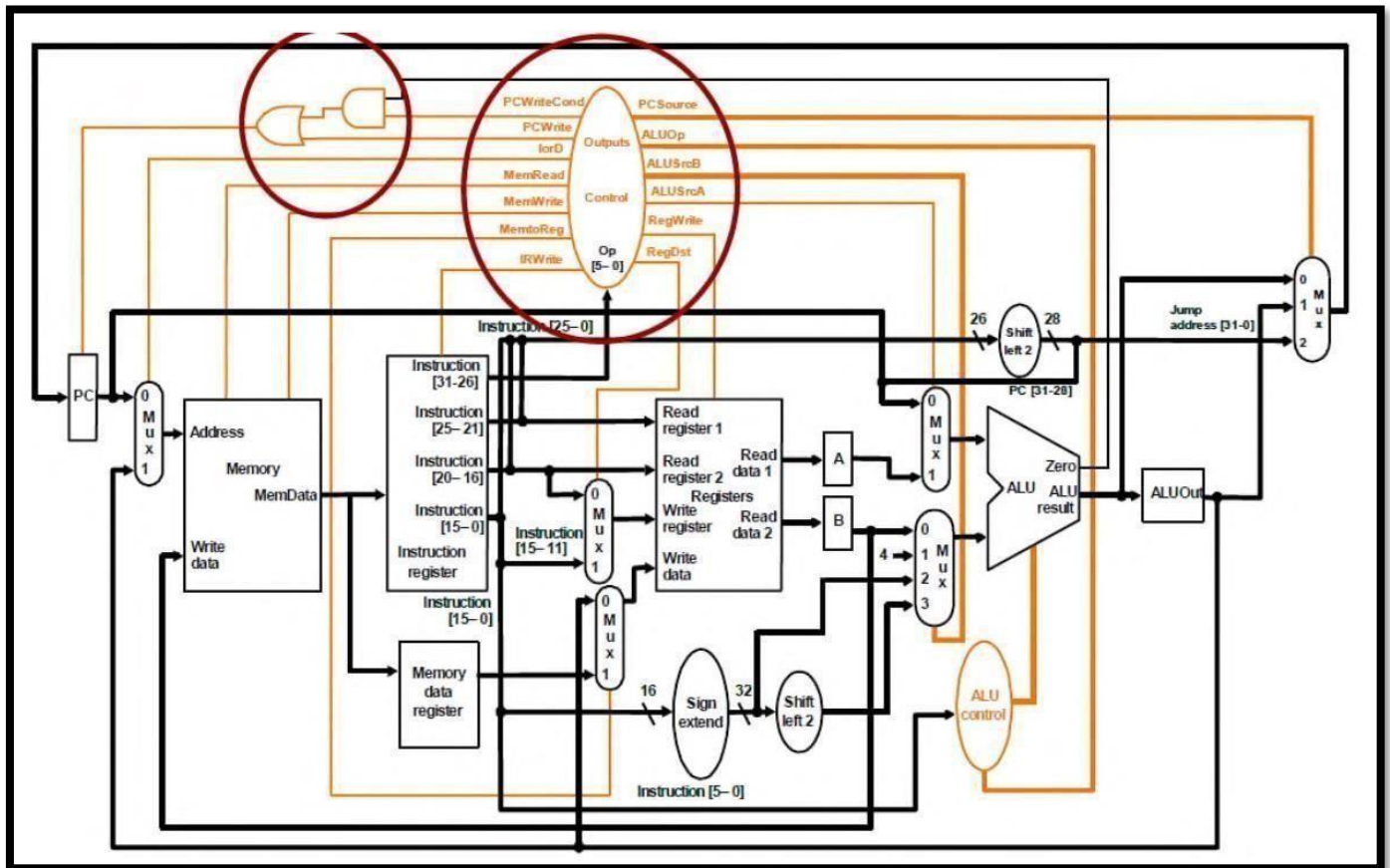
Q4) Show how the Instructions in the sequence given below will proceed through the pipeline using forwarding method :

**** Hint : we will predict that the beq instruction isn't taken.**

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
beq \$1,\$2,X	IF	ID	EX	ME	WB									
lw \$10,0(\$11)		IF	ID	EX	ME	WB								
sub \$14,\$10,\$10			IF	stall	ID	EX	ME	WB						
X: add \$4,\$1,\$2					IF	ID	EX	ME	WB					
lw \$1,0(\$4)						IF	ID	EX	ME	WB				
sub \$1,\$1,\$1							IF	stall	ID	EX	ME	WB		
Add \$1,\$1,\$1									IF	ID	EX	ME	WB	

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Q5) Show the Signal in The Cycle Number 1 in Multicycle Datapath:



Signal	Value
PCWrite	1
lorD	0
MemRead	1
MemWrite	0
IRWrite	1
PCSource	00
ALUOp	00
ALUSrcB	01
ALUSrcA	0
RegWrite	0

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Q6) Assume MIPS Big Endian for the Byte Order Convention in this Problem. The Following Figure Shows the content trunk of the memory with the corresponding addresses.

**** Hint :** Assume the value represented in the memory are signed and in 2's complement form.

**** The Value Of Register \$s0 = 0 , \$s1 = 4 and \$s2 = 9**

1. What's the content of registers \$t0,\$t1,\$s1,\$s2 after executing the program ?
2. Show The content of the same trunk of memory after executing the program ?

lw \$t0,0(\$s1)

lw \$t1,-9(\$s2)

mult \$t1,\$t0

mflo \$s1

sw \$s1,0(\$s0)

mfhi \$s2

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(Solutions)

Before :

Address	Content
0x7	0x01
0x6	0x00
0x5	0x00
0x4	0x00
0x3	0x0d
0x2	0x00
0x1	0x00
0x0	0x00

After :

Address	Content
0x7	0x01
0x6	0x00
0x5	0x00
0x4	0x00
0x3	0x0d
0x2	0x00
0x1	0x00
0x0	0x00

\$t0	0x00000001
\$t1	0x0000000d
\$s1	0x0000000d
\$s2	0x00000000

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Q7) Compute the following information about ram executing in two Computer (Computer 1 & Computer 2) and find performance ISA.

**** Hint :** السؤال صيغته مش 100% صححة لكن المدهم النكرة انك تقارن فرق السرعة بين الاثنين

	Computer 1	Computer 2
IC	$2.0 * 10^9$	$3.0 * 10^9$
CR	1 GHz	2 GHz
CPI	2	4

- Find Which Computer Is Faster , Then Calculate How Much Is Faster Than Another Computer ?

$$\text{Execution Time Computer 1} = (2 * 10^9) * (2) * (1 / 1 * 10^9)$$

$$\text{Execution Time Computer 1} = 4 \text{ seconds}$$

$$\text{Execution Time Computer 2} = (3 * 10^9) * (4) * (1 / 2 * 10^9)$$

$$\text{Execution Time Computer 2} = 6 \text{ seconds}$$

$$\text{Speed Up} = 6 / 4 = 1.5$$

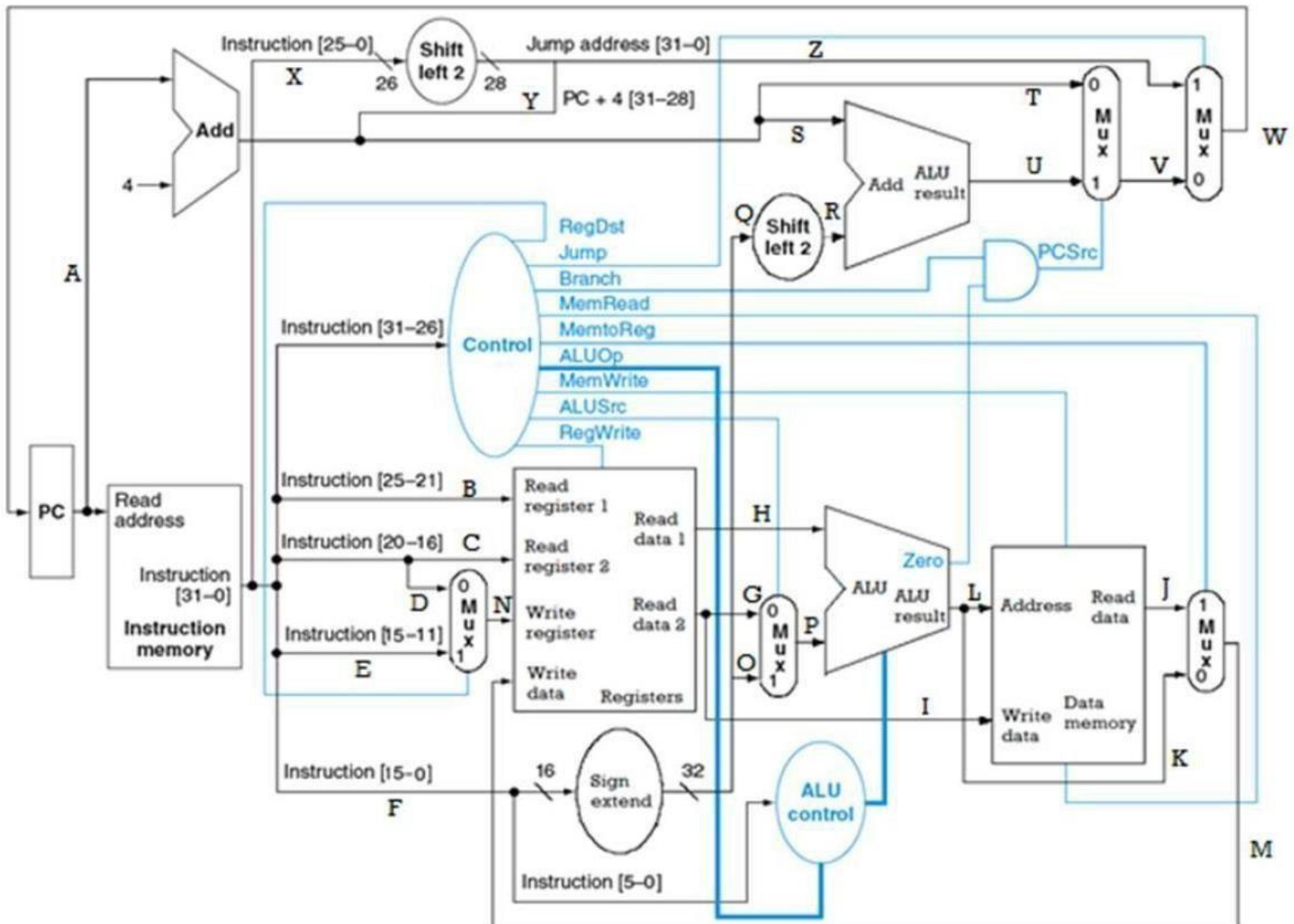
Computer 1 Faster Than Computer 2 By 50%

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Computer Architecture and Organization Final Exam (Solutions)

Q8) Consider the following MIPS single cycle processor, and the snapshot of the instruction memory, data memory and the register file.

Execute the instructions and fill the following table (Buses and Control Signals), assuming that the PC register contains 0x00400004.



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(Solutions)

Instruction Memory		
Address	Contents	Instruction
0x00400000	0x0253A825	OR \$S5, \$S2, \$S3
0x00400004	0x02309020	ADD \$S2, \$S1, \$S0
0x00400008	0x12530001	Beq \$S2, \$S3, 1
0x0040000c	0x8E93FFFE	LW \$S3, -2(\$S4)
0x00400010	0x08100001	J 0x100001

Register File	
Register#	Data
16 (\$s0)	0x0000000A
17 (\$s1)	0x00000001
18 (\$s2)	0x02E40160
19 (\$s3)	0x00000011
20 (\$s4)	0x00020006
21 (\$s5)	0x00012304

Data Memory	
Address	Data
0x00020002	0x13
0x00020003	0x2E
0x00020004	0xA4
0x00020005	0xFF
0x00020006	0x08
0x00020007	0x2C

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Instr. Buses	ADD \$S2, \$S1, \$S0	Beq \$S2, \$S3, 1	LW \$S3, -2(\$S4)	J 0x100001	ADD \$S2, \$S1, \$S0
A	0x00400004	0x00400008	0x0040000c	0x00400010	0x00400004
B	17 (\$s1)	18 (\$s2)[	20 (\$s4)	---	
C	16 (\$s0)	19 (\$s3)	---	---	
D	16 (\$s0)	---	19 (\$s3)	---	
E	18 (\$s2)	-----	19 (\$s3)	---	
F	----	0x0001	0x ff fe	---	
G	0x0000000A	0x00000011	----	---	
H	0x00000001	0x0000000B	0x00020006	---	
I	-----	---	----	---	
J	-----	---	0xA4FF082C	---	
K	0x0000000B	----	----	---	
L	0x0000000B	0x FF FF FF FA	0x00020004	---	
M	0x0000000B	----	0xA4FF082C	---	
N	18 (\$s2)	---	19 (\$s3)	---	
O	-----	-----	0x FF FFff fe	---	
P	0x0000000A	0x00000011	0x FF FFff fe	---	
Q	-----	0x0000 0001	-----	---	
R	----	0x0000 0004	----	---	
S	-----	0x0040000c	0x00400010	0x00400014	
T	0x00400008	0x0040000c	0x00400010	0x00400014	
U	-----	0x00400010	---	---	
V		0x0040000c	0x00400010	---	
W	0x00400008	0x0040000c	0x00400010	0x00400004	
X	----	----	---	00b0x10 00 01	
Y	----	----	---	0	
Z	-----	----	----	0x00400004	
Control Sig.					
RegDst	1	X	0	X	
RegWrite	1	0	1	0	
ALUSrc	0	0	1	X	
MemRead	0	0	1	0	

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MemWrite	0	0	0	0	
MemtoReg	0	X	1	X	
Jump	0	0	0	1	
Branch	0	1	0	0	
ALUOpr1	1	0	0	1	
ALUOpr0	0	1	0	1	
PCSrc	0	0	0	X	
Registers					
16 (\$s0)	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	
17 (\$s1)	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	
18 (\$s2)	0x0000000B	NO CHANGE	NO CHANGE	NO CHANGE	
19 (\$s3)	NO CHANGE	NO CHANGE	0xA4FF082C	NO CHANGE	
20 (\$s4)	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	
21 (\$s5)	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	

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Q9) Consider Two Different Implementations, M1 and M2, of the same Instruction set. There are three classes of Instruction (A,B and C) in the instruction set. M1 has a Clock Rate of 80 MHz and M2 has a clock rate of 100 MHz, The Average Number of cycles for each instruction class and their frequencies (for a typical program) are as follows:

Instruction Class	M1 – Cycles/Instruction Class	M2 – Cycles/Instruction Class	Frequency
A	1	2	60%
B	2	3	30%
C	4	4	10%

A) Calculate the average CPI for each machine, M1 and M2.

B) Calculate the average MIPS rate for each machine, M1 and M2.

Solution : a)

For machine M1:

$$\begin{aligned}\text{Clocks per instruction (CPI)} &= (60/100)*1 + (30/100)*2 + (10/100)*4 \\ &= 0.6 + 0.6 + 0.4 \\ &= 1.6\end{aligned}$$

For machine M2:

$$\begin{aligned}\text{Clocks per instruction (CPI)} &= (60/100)*2 + (30/100)*3 + (10/100)*4 \\ &= 1.2 + 0.9 + 0.4 \\ &= 2.5\end{aligned}$$

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(Solutions)

b)

For machine M1:

$$\begin{aligned}\text{Average MIPS rating} &= \text{Clock Rate} / (\text{CPI} * 10^6) \\ &= (80 * 10^6) / (1.6 * 10^6) \\ &= 50\end{aligned}$$

For machine M2:

$$\begin{aligned}\text{Average MIPS rating} &= \text{Clock Rate} / (\text{CPI} * 10^6) \\ &= (100 * 10^6) / (2.5 * 10^6) \\ &= 40\end{aligned}$$

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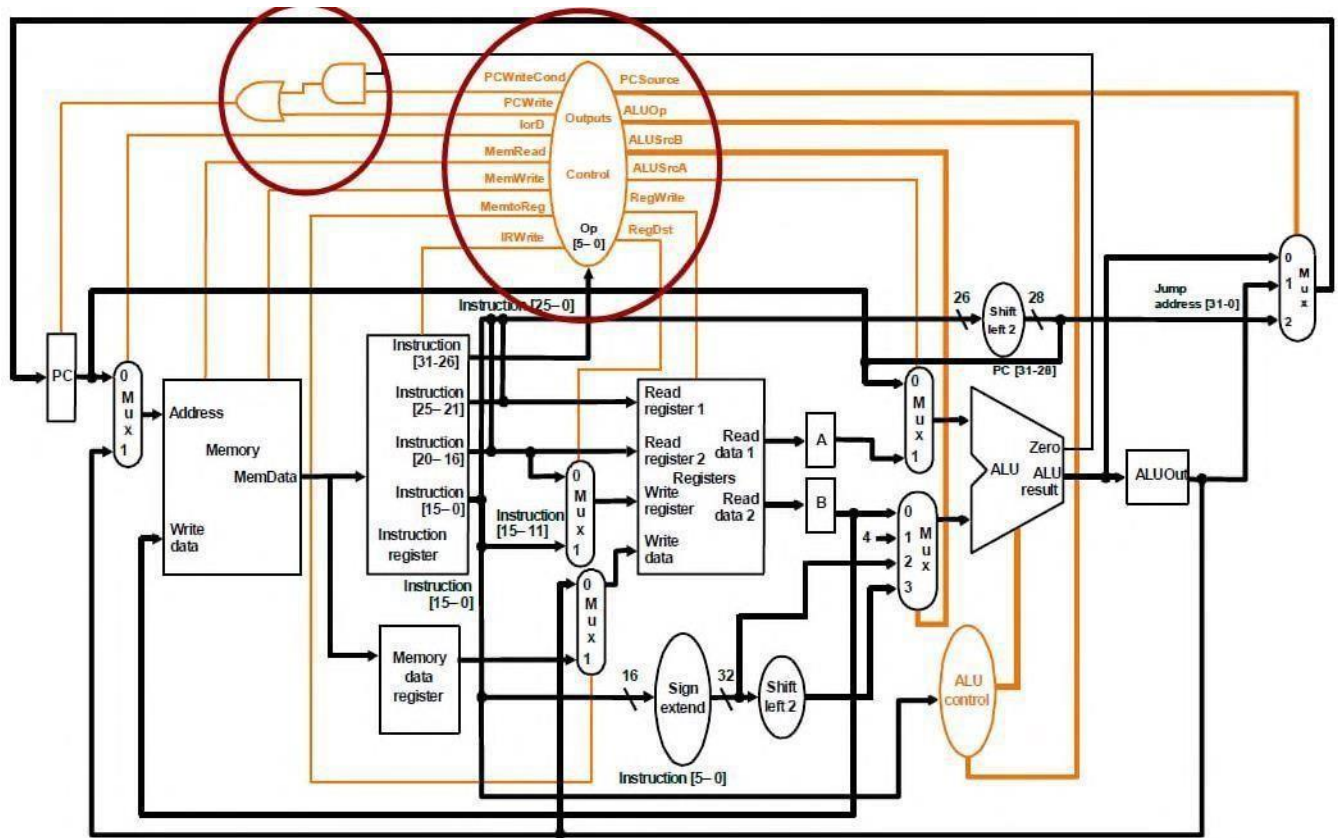
(Solutions)

Q10) Find The Value of \$s2 Register After Execution the following Instruction, Hence the Value of \$s0 = 0x00000010 and \$s1 = 0x00000020.

Instruction:	Solution:
and \$s2,\$s1,\$s0	\$s2 = 0x00000000
or \$s2,\$s1,\$s0	\$s2 = 0x00000030
slt \$s2,\$s0,\$s1	\$s2 = 0x00000001
sll \$s2,\$s0,2	\$s2 = 0x00000040
addi \$s2,\$s0,30	\$s2 = 0x0000002E

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Q11) In the Multicycle Datapath Draw or Show the lines in (Beq) Instruction for 3 Cycle, Add Control line If you need to.



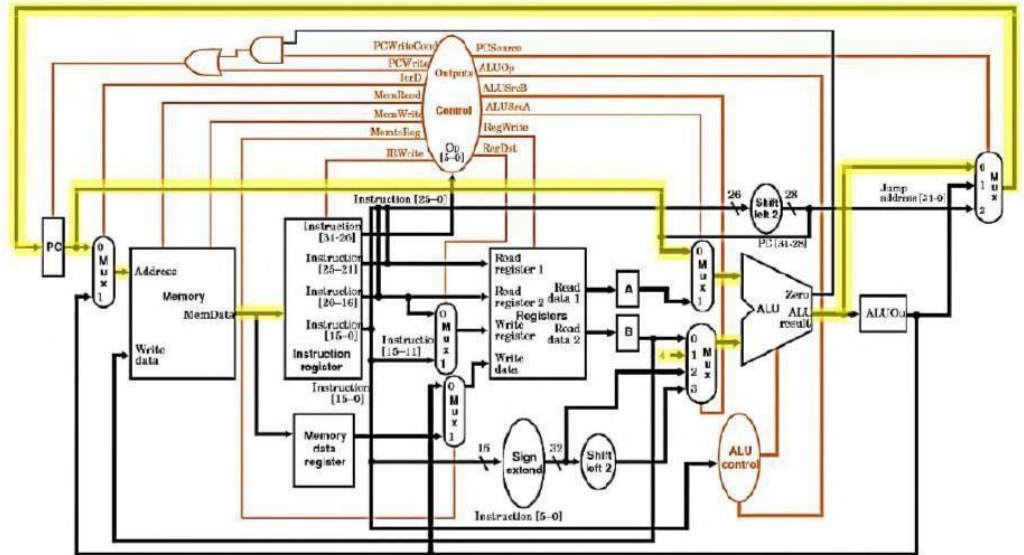
Solution : (see next pages)

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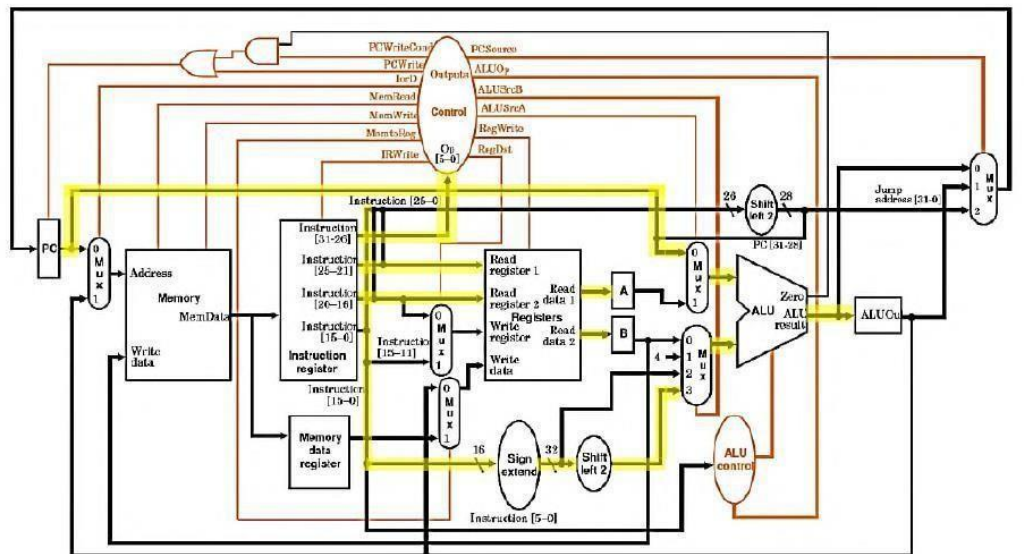
BRANCH: CYCLE 1

Signal	Value
PCWrite	1
lorD	0
MemRead	1
MemWrite	0
IRWrite	1
PCSource	00
ALUOp	00
ALUSrcB	01
ALUSrcA	0
RegWrite	0



BRANCH: CYCLE 2

Signal	Value
ALUOp	00
ALUSrcB	11
ALUSrcA	0

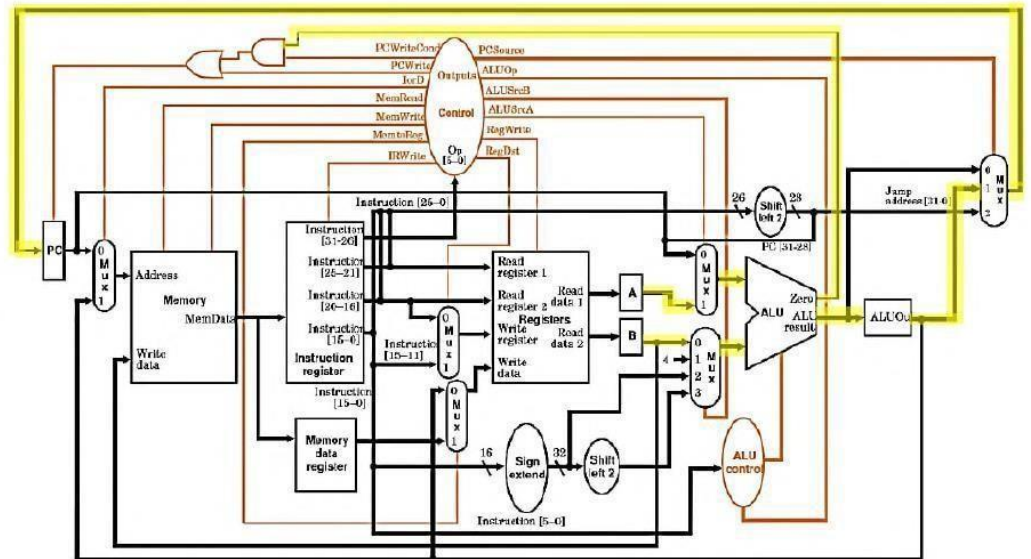


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BRANCH: CYCLE 3

Signal	Value
ALUOp	01
ALUSrcB	00
ALUSrcA	1
PCSource	01
PCWriteCond	1

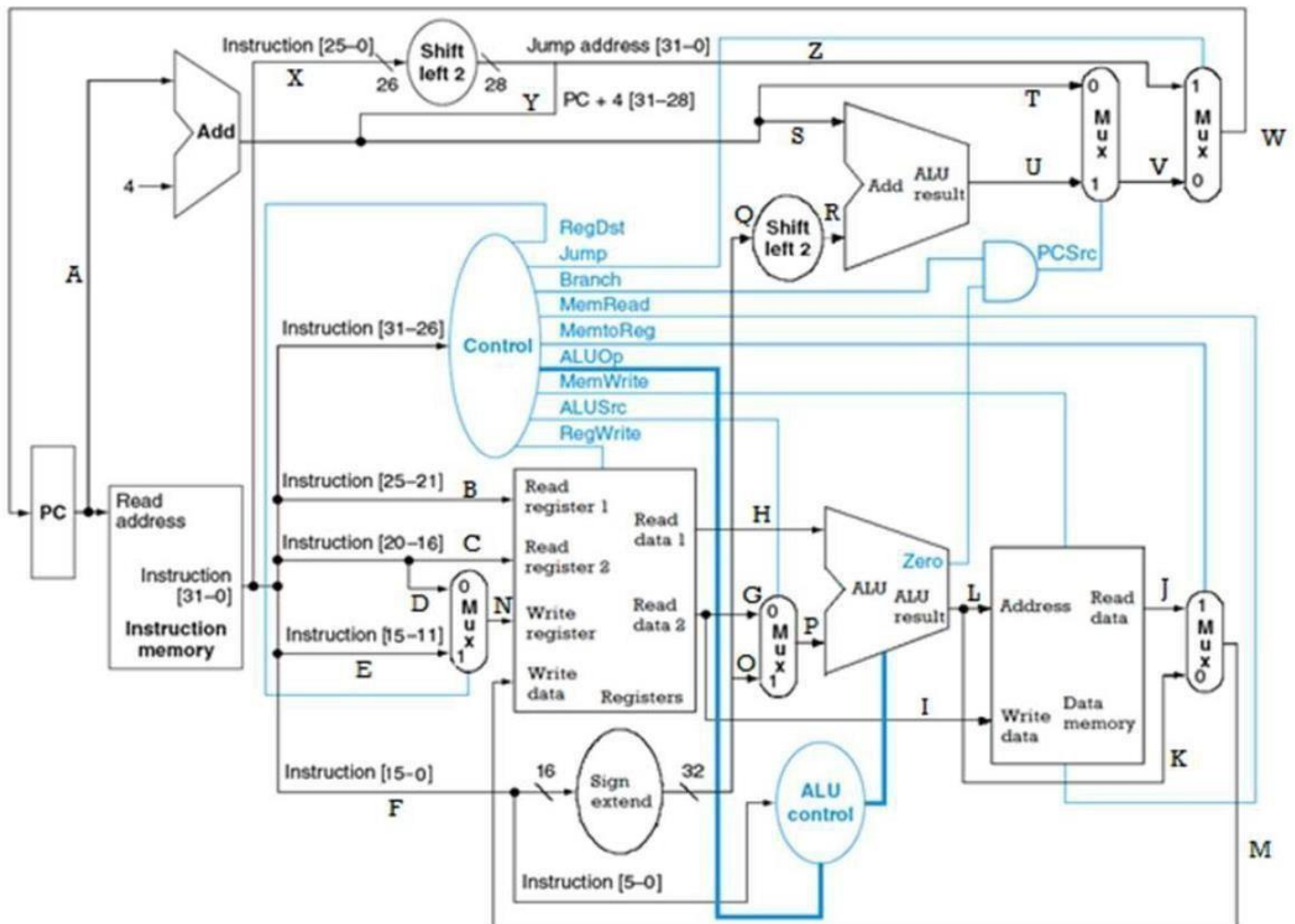


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(Solutions)

Q12) Consider the following MIPS single cycle processor. Find The Control Signal For SW (Store) and ADDI (Add Immediate) Instructions.



Instr.	RegDst	RegWrite	ALUSrc	MemRead	MemWrite	MemtoReg	Jump	Branch	ALUOp1	ALUOp0	PCSrc
SW ...	X	0	1	0	1	X	0	0	0	0	0
addi ...	0	1	1	0	0	0	0	0	1	1	0

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